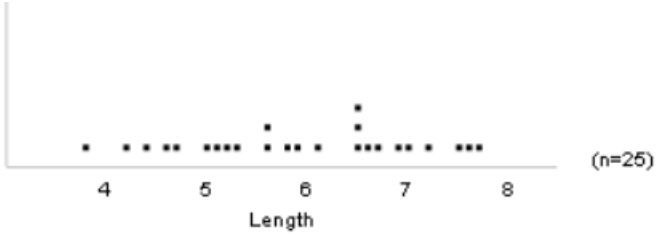
Solutions to Exploration 2.1A

1. (one option, will vary by student) Score, engaged, larger, living, devotion, freedom, government, people, world, liberty
2.

Word	Length (# letters)	Contains e? (Y/N)
Score	5	Yes
Engaged	7	Yes
Larger	6	Yes
Living	6	No
Devotion	8	Yes
Freedom	7	Yes
Government	10	Yes
People	6	Yes
World	5	No
Liberty	7	Yes

3. Obs units = each word, variables = length, contains e
4. Length of word = quantitative
5. 6.7, statistic (because it's based on the sample of 10 words), $\bar{x}$
6. Parameter (because it's based on the population of all words), μ
7. 8/10, statistic (because it's based on the sample of 10 words), $\hat{p}$
8. Parameter (because it's based on the population of all words), π
9. No, the words selected are not representative. In general, the values of the statistics should be close to the values of the parameters
10. Here is an example from a class of 25 students



a. The value of the statistic average word lengths on a single students sample

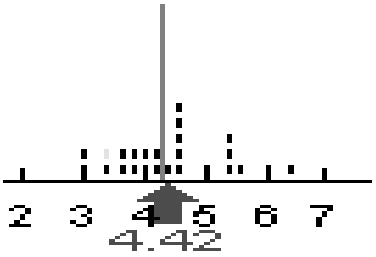
b. Centered at 6 or 7 roughly symmetric, varies from between about 4 and 8 letters on average

11. a. 23/25 (from dotplot above) b. 19/25 (raw data not shown)

12. Biased. Class tends to overestimate length of words and proportion of e words
13. Representativeness was based (mainly) on capturing the literary "essence" of the Gettysburg address, not on word length
14. No, the same problem would have resulted---people would still have a tendency to ignore shorter words.
15. No, biased towards longer words--they're more likely to be selected because they are longer
16. Put each word on the same sized slip of paper, put the slips in a hat and draw one out
17. 5 random numbers generated vary. (one option, yours will be different)

Word	Length (# letters)	Contains e? (Y/N)
Four	4	No
This	4	No
Dedicated	9	Yes
Now	3	No
We	2	Yes

18. a. 4.4 average word length, 0.4 proportion of e words,
    b. Here is an example from one class of 25 students



    c. this distribution is centered lower than the one produced from non-random sampling
19. a. varies, b varies, c. Yes, appears unbiased since (on average) statistics are getting close to parameter
21. varies … one option presented here: 4.4 (first), 5.7 (second), 5.05 (average)
22. varies, one option is 4.43
23. Bell-shaped curve, centered at approximately 4.29 (the value of the population mean). Ranges from about 3 to 6.5, sd is about 0.95
24. Shape is similar, Mean is the same, sd is smaller (0.4730 now, was 0.95 before)
25. No, similar center because unbiased sampling. Biased sampling would not remove the bias.

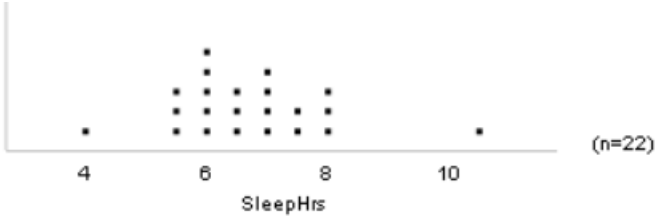
Solutions to Exploration 2.1B

1. Population: All Ohio voters, Sample: 1,421 Ohio voters who took the survey
2. Yes, because it was randomly selected
3. Statistic, it is a numerical summary of the sample. $\hat{p}$
4. Long-run proportion of all Ohio voters (population) who think the bill is a good idea (π)
5. No, the statistic is likely close to the value of the parameter, but not exactly the same
6. Each trial represents a randomly selected voter; The number of trials is 1,421; probability of success is 0.50
7. 50%
8. 50% of 7.8 million is 3.9 million.
9. 3,899,999/7,799,999 is essentially 50%
10. Null: 50% of Ohio voters are in favor of the ban, Alt: More than 50% of Ohio voters are in favor of the ban; 1,421 tosses, 1,000 repetitions. p-value of approximately 0
11. The probability of getting 55% or more of a sample to be in favor of the bill when the null hypothesis is true is approximately zero. We have strong evidence that the null hypothesis is wrong.
12. The SD on the applet is 0.014, which is essentially the same as what the formula yields (0.013)
13. Yes, there are at least 10 successes and 10 failures (55% of 1,421 and 45% of 1,421 are both > 10) and the population size (7.8 million) is substantially larger than the sample size (1,421).
14. The standardized statistic is 3.77, which means our sample proportion on 0.55 is 3.77 standard deviations about the theoretical mean (0.50) of the null distribution.
15. The p-value from the normal approximation is 0.0001. Yes, they are both extremely small.

Solutions to Exploration 2.2

1. Population: All students at your school, Parameter: population average amount of time all students at your school slept last night, symbol: μ
2. Null hypothesis: The average length of time all students at your school slept last night (μ) was 8 hours.

   Alt hypothesis: The average length of time all students at your school slept last night was less than 8 hours.
3. Take a simple random sample by getting a listing of all students at your school and using a random device to randomly choose some.
4. The number of hours slept last night
5. (Various answers are acceptable) Statistics classes often represent many different majors and/or class standings.
6. (Various answers are acceptable) The time of day your class meets may be a way your class is not a good representation (if your class is in the morning, you may tend to have students who got less sleep; if in the afternoon, you may tend to have students who got more sleep).
7. Biased, Unbiased
8. (Your answer will vary). We'll use 7.
9. (Various answers are acceptable). We used hours.
10. (Various answers; this is the dataset posted on the textbook data page from one of the authors' classes)



    Fairly symmetric

11. 6.705 hours, x-bar;
12. 10
13. 10/22 = 45%
14. Not very skewed, because if it was skewed a high or low percentage of data values would be above the mean.

15. 7
16. 6.5
17. No, the mean (6.705) and median (6.5) are similar. Yes, they are preliminary evidence in favor of the alternative hypothesis because they are both a fair bit below 8.
18. SD=1.297
19. Sleep times, students in my class, 1.297 hours, on average, students in my class, 1.297 hours, 6.705.
20. There is at least one (10.5 hours), and maybe another (4 hours) which are outside the bulk of the data; most students slept between 5.5 and 8 hours.
21. Mean = 7.136. Gotten larger because mean is impacted by extreme values. Median is still 6.5, hasn't changed because median not impacted much by extreme values. Standard deviation is now 3.036, which is larger than before because the SD is impacted by extreme value.
22. Mean, Standard Deviation
23. 6.705
24. (Answers will vary) 8.02
25. A. Bell shaped curve, b. 8—that is the population mean, c. (Answers will vary) SD = 0.32
26. $p < 0.001$. We have very strong evidence that the population average sleep time at our school was less than 8 hours last night
27. (6.705 - 8)/0.32 = -4.05
28. We have very strong evidence that the population average sleep time at our school was less than 8 hours last night.
29. Yes, more than 20 observations and the data is not strongly skewed; in fact it is fairly symmetric.
30. $\frac{1.297}{\sqrt{22}} = 0.2765$, had 0.32 earlier. Similar because the standard deviation of the sample means is estimated by simulation as 0.32, and predicted to be 0.2765. This is a version of the Central Limit Theorem.
31. $t = \frac{6.705-8}{1.297/\sqrt{22}} = -4.68$
32. We have very strong evidence that the population average sleep time at our school was less than 8 hours last night
33. Yes, does a very good job predicting the distribution. This is because the validity conditions are met.
34. Approximate (simulation) p-value using t-statistics is 0.000

35. Theory-based (predicted) p-value using t-statistics is 0.0001
36. Because the validity conditions are met.
37. (Answers will vary) No, not completely. There are some things about the class that are similar to all students at our school and other things that aren't. Taking a random sample would be better
38. Biggest limitation is the way the data was collected (convenience sample)
39. A random sample would be much better.
40. A. Population is skewed, but still has a mean of 8 and SD of 1.5. B. Took 1000 random samples from this population and found proportion of times that got a sample mean less than 6.705. p-value is < 0.001. Strong evidence population mean sleep time at our school last night was less than 8 hours (could also use *t*-statistics)
41. Yes, the simulated null distribution is bell shaped and centered at 8, which makes sense since the sample size is larger than 20 and the distribution is symmetric
42. The change in population shape had virtually NO impact on the distribution of sample means
43. The change in population shape had virtually no impact on the strength of evidence (still very strong evidence against null)
44. Null: The population median sleep hours last night was 8 hours, Alt: The population median sleep hours last night was less than 8 hours
45. A. Sample median = 6.5, B. Take 1000 random samples from a population with median of 8 and compute the sample median each time, C. compare 6.5 to the 1,000 random medians and see what proportion of the time random medians were less than the observed median (6.5)
46. Very strong evidence against the null hypothesis
47. No, because the median is not 8.

Solutions to Exploration 2.3

1. 25%
2. Null: Long-run proportion of times that psychic will pick correct target is 25%;
   Alt: Long-run proportion of times that psychic will pick correct target is more than 25%
3. 709/2124 = 0.334. Yes, larger than null hypothesized value
4. The p-value is approximately 0.
5. The theory-based approach gives a standardized Z statistic of 8.92 and a p-value of approximately 0. The validity conditions are met because the number of "hits" is 709, and the number of misses is 1,415--both greater than 10.
6. Very strong
7. Yes, because the evidence is very strong.
8. (Answers will vary) Somewhere between 1 and 10% is typical.
9. Yes, the p-value is less than 0.05
10. Smaller significance level if they require stronger evidence to be convinced
11. Could have made a Type I error (false alarm) because the researchers rejected the null hypothesis. More specifically, the researchers said there was very strong evidence that the psychics did better than guessing, but they could have, in fact, just been guessing.
12. The p-value is approximately 0.21 (one sided test), which cannot be rejected at the 0.05 or 0.10 level. We don't have evidence, from this study alone (50 sessions), that the long-run proportion of times that psychic will pick the correct target is larger than 25%. The p-value (0.21) is larger than a sig level of both 0.05 and 0.10.
13. Could be a Type II error (missed opportunity). That is, our conclusion is that we don't have evidence the long-run probability is larger than 25%, when, in fact, it is larger than 25%.
14. Reduces the chances of a Type I error (false positive) because you make it very hard to reject the null hypothesis.
15. But, the consequence is that the probability researchers find significant evidence of psychic ability is much lower, even if their true probability is larger than 25% because you've made it very hard to reject the null hypothesis.
16. Alt hypothesis. Type II error is possible
17. 22.5%. Increasing the sample size can improve the power of the test, another option is increasing the significance level to 10%.